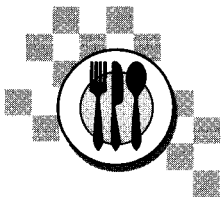
CHAPTER 3



Sources of Food Contamination

About This Chapter

In this chapter you will learn:

1. Where microorganisms in foods come from
2. The difference between foodborne infection and foodborne intoxication
3. How microbes in food cause illness in people
4. How specific foods and ingredients become infected
5. How contamination of foods can be prevented and controlled

Introduction

Nutrients for Microorganisms

Food products provide nutrients that people need. Microorganisms need many of the same nutrients, so food products are an ideal source of nutrition for microorganisms. Foods also generally have a pH value (acidity level) that encourages growth of microorganisms.

Where Do Microorganisms in Food Come From?

Microorganisms are everywhere: in soil, feces, air, and water. Food can be contaminated any time it comes in contact with these substances during harvesting, processing, distribution, and preparation.

Animal carcasses. The intestines of animals used for meat are full of microorganisms, and even animals that seem healthy may have microorganisms in their liver,

kidneys, lymph nodes, and spleen. It is easy for these microorganisms to reach the meat during slaughter and butchering. As the meat is cut into retail portions, more and more surface area is exposed to the microorganisms.

Controlling Growth of Microbes

Food processors try to control the growth of these microorganisms by controlling the temperature (refrigeration) and oxygen and moisture levels (wrapping and packaging) during processing and distribution of foods.

Temperature. Refrigeration is one of the most important ways contamination of foods is controlled. Refrigeration prevents outbreaks of foodborne illness by slowing the growth of microbes. But when foods are not handled properly in cold storage, they can become heavily contaminated. The growth rate of microorganisms can increase tremendously with only a slight increase in temperature.

Food volume and container size. Foods tend to cool slowly in air, and large containers or large volumes of food take a long time to cool through to the center. The surface of the food can feel well chilled while the middle is still warm. Slow cooling of large pieces of meat or large containers of broth in the refrigerator has caused many outbreaks of foodborne illness caused by growth of *Clostridium perfringens.*

Infection vs. Intoxication

There are two types of foodborne illness caused by microorganisms: infection and intoxication.

- Foodborne *infection* occurs when the microorganism is eaten with the food and multiplies in the person's body. Examples include *Salmonella, Shigella,* and some types of enteropathogenic *Escherichia coli.*
- Foodborne *intoxication* occurs when microorganisms grow and release toxins into the food before it is eaten. When the food is eaten, the toxins cause illness. Examples include *C. perfringens* and some strains of enteropathogenic *E. coli.*

THE CHAIN OF INFECTION

A small number of harmful microorganisms (pathogens) in food are unlikely to cause a foodborne illness. However, these few pathogens could cause a problem under certain conditions. One model that illustrates how various factors and events can be linked together to cause an infection is called the "chain of infection."

A chain of infection is made up of four links: agent, source, mode of transmission, and host. To cause a foodborne infection, each of these links must be present in the environment in which the food is produced, processed, or prepared.

- The *agent* is the pathogen that causes the illness.

- The *source* is where the pathogen comes from.
- The *mode of transmission* is how the pathogen is carried from the source to the food.
- The *host* is the food that supports growth of the pathogen.

To survive and grow, the pathogen also needs nutrients, moisture, the right level of acidity or alkalinity (pH) and oxidation-reduction potential, and lack of competitive microorganisms and inhibitors. The food also needs to be held in the best temperature range for that organism long enough to allow it to grow enough to cause infection or intoxication.

The chain of infection model shows that foodborne diseases have many causes. A disease agent (pathogen) has to be present, but each of the other steps is also essential in causing foodborne disease.

HOW FOODS BECOME CONTAMINATED

The food itself is the most common source of contamination. Equipment and waste products are other common sources.

Dairy Products

The udders of cows and milking equipment can contaminate milk products, although equipment with well-designed sanitary features and control of disease in dairy cows have made dairy products more wholesome. Pasteurization of milk products in processing plants has also reduced pathogens in milk. However, dairy products can be cross-contaminated by items that have not been pasteurized. Cross-contamination occurs when utensils or equipment used for unpasteurized milk are used for pasteurized milk, or when staff working with unpasteurized milk products move to an area containing pasteurized milk products. Not all dairy products are pasteurized, so some pathogens (especially *Listeria monocytogenes*) have become more common in the dairy industry. (See Chap. 11 for more information.)

Red-Meat Products

The muscles of healthy animals are nearly free of microorganisms while alive. Meat is contaminated by microorganisms on the animal's surfaces that have external contact (hair, skin, intestines, and lungs). While the animal is alive, its white blood cells and antibodies control infection. But these defense mechanisms are lost during slaughter.

Microorganisms first reach the meat if contaminated knives are used to bleed animals. The blood is still circulating and quickly carries these microorganisms throughout the animal's body. Microorganisms reach the surface of the meat when it is cut, processed, stored, and distributed. Meat can also be contaminated if it comes in contact with the hide, feet, manure, dirt, and visceral (intestinal) contents if the digestive organs are punctured. (See Chap. 12 for more information.)

Poultry Products

Poultry may be contaminated by *Salmonella* and *Campylobacter* during processing. These microorganisms are easily spread from one carcass to another during defeathering and removal of the intestines (evisceration). *Salmonellae* can also be transferred from contaminated hands, gloves, and processing tools. (See Chap. 12 for more information.)

Seafood Products

Seafoods may be contaminated with microbes during harvesting, processing, distribution, and marketing. Seafoods are excellent sources of proteins, B vitamins, and a number of minerals that bacteria need to grow. Therefore, microbes grow well on or in seafoods. Seafoods are handled a lot from the time they are harvested until they are eaten, which provides many opportunities for contamination. They may be also sometimes be stored without being refrigerated, which allows microorganisms to grow. (See Chap. 13 for more information.)

Ingredients

Ingredients (especially spices) can carry harmful or potentially harmful microorganisms and toxins. The amounts and types of these microbes and toxins depends on where and how the ingredient was harvested and how the ingredient was processed and handled. The food plant management team needs to know the type of hazards that can occur with each ingredient. Food processors should only obtain materials from suppliers that use good practices.

OTHER SOURCES OF CONTAMINATION

Equipment

Equipment can be contaminated during production and while it is not being used. Most equipment is designed to be hygienic, but it can still collect microorganisms and other debris from the air, employees, and food ingredients. Food is less likely to be contaminated if equipment is designed to be hygienic and is cleaned regularly and thoroughly.

Employees

The most common source of microorganisms in foods is employees. The hands, hair, nose, and mouth carry microorganisms that can be transferred to food during processing, packaging, preparation, and service by touching, breathing, coughing, or sneezing. Because the human body is warm, microorganisms grow and multiply rapidly. Therefore, sanitary practices, such as good handwashing and use of hairnets and disposable plastic gloves, are essential. (See Chap. 4 for more information.)

Air and Water

Water is used for cleaning and as an ingredient in many processed foods. However, if the water is not pure, it can contaminate foods. If the water source is contaminated, another source should be used, or the water should be treated by chemicals, ultraviolet units, or other methods.

Microorganisms in the air can contaminate foods during processing, packaging, storage, and preparation. The best ways to reduce air contamination are to use filters for air entering food-processing and preparation areas and to package or cover food products to reduce contact with air.

Sewage

Raw, untreated sewage can carry microorganisms, causing typhoid and paratyphoid fevers, dysentery, and infectious hepatitis. Raw sewage may contaminate food and equipment through faulty plumbing. If raw sewage drains or flows into drinking-water lines, wells, rivers, lakes, and ocean bays, the water and seafood will be contaminated. To prevent this kind of contamination, toilet facilities and septic tanks should be separated from wells, streams, and other water sources. Raw sewage should not be used to fertilize fields where fruits and vegetables are grown. (See Chap. 8 for more information.)

Insects and Rodents

Food and food waste attract flies and cockroaches to kitchens, foodservice operations, food-processing facilities, toilets, and garbage. These insects transfer dirt from contaminated areas to food through their waste products; mouth, feet, and other body parts; and saliva. Any pests should be eradicated and prevented from entering food-processing, preparation, and serving areas.

Rats and mice carry dirt and disease with their feet, fur, and feces. They transfer dirt from garbage dumps and sewers to food or food-processing and foodservice areas. (See Chap. 9 for more information.)

HOW TO PREVENT AND CONTROL CONTAMINATION OF FOODS

The Environment

Foods should not be touched by human hands if they will be eaten raw or after they have been cooked. If contact is necessary, workers should thoroughly wash their hands before handling the food and wash regularly during handling or use disposable plastic gloves. During storage, holding, and service, processed and prepared foods should be covered with a clean cover that fits well and will not collect loose dust, lint, or other debris. If the food cannot be covered, it should be held in an enclosed dust-free cabinet. Foods in single-service wrappers or containers, such as milk and juice, should be served directly from these containers. Foods on a buffet

should be served on a steam table or ice tray and protected by a transparent shield from sneezes, coughs, and other contact with air, employees, and customers. Food that touches an unclean surface should be thoroughly washed or thrown away. Equipment and utensils used in food processing, packaging, preparation, and service should be cleaned and sanitized between use. Foodservice managers should train employees to handle dishes and eating utensils so that their hands do not touch any surface that will come into contact with food or the consumer's mouth.

During Storage

Storage facilities should have plenty of space and an organized storage layout, and stock should be rotated. These measures help to reduce contamination from dust, insects, rodents, and dirt, and allow for easy cleaning. Storage area floors should be swept or scrubbed, and shelves or racks should be cleaned and sanitized. (See Chaps. 5, 6, and 7 for more information.) Trash and garbage should not be allowed to accumulate in food storage areas.

Litter and Garbage

The food industry generates large amounts of waste from used packaging, containers, and waste products. Refuse should be kept in appropriate containers and removed from the food area regularly. The best method (required by some regulatory agencies) is to use separate containers for food waste from those used for litter and rubbish. Clean, disinfected, seamless trash containers should be kept in all work areas. All containers should be washed and disinfected daily; plastic liners may also be used as a cheap way to keep trash. Close-fitting lids should be kept on the containers, except when they are being filled and emptied. Containers in food-processing and food preparation areas should not be used for litter and rubbish from other areas.

Toxic Substances

Poisons and toxic chemicals should not be stored near food products. Only chemicals required for cleaning should be stored in the building, and these should be clearly labeled. Only cleaning compounds, supplies, utensils, and equipment approved by regulatory or other agencies should be used in food handling, processing, and preparation.

SUMMARY

- Food products are attractive breeding sites for microorganisms because they are rich in nutrients.
- Most microorganisms come from water, air, dust, equipment, sewage, insects, rodents, and employees.
- The chain of infection is a model that shows how foodborne diseases are carried.

- Raw materials can also be contaminated from the soil, sewage, live animals, external surfaces (skin, shells, etc.), and internal organs of meat animals. With modern health care, diseased animals rarely cause illness.
- Chemicals can contaminate foods by accidental mixing.
- Good housekeeping, sanitation, storage, and garbage-disposal practices prevent and control contamination of food.

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STUDY QUESTIONS

1. What is the difference between foodborne infection and foodborne intoxication?
2. Give three reasons why seafood is easily contaminated.
3. What is the most common source of contamination of food?
4. List two ways that foods can be protected from microbes on hands.
5. How can food be protected against microbes in air?
6. Describe three features of buffet serving areas that protect the food from microbial contamination.
7. Where and how should garbage or trash be stored?

TO FIND OUT MORE ABOUT FOOD CONTAMINATION

1. Call the USDA Meat and Poultry Hotline, (1-800) 535-4555. Ask for information about keeping meat and poultry free from contamination.
2. Call the U.S. Environmental Protection Agency Office of Drinking Water SAFE DRINKING WATER HOTLINE, (1-800) 426-4791. Ask for information about keeping water supplies free from contamination.

3. Look in the telephone directory and call your local water utility for information on how contaminants are controlled in your local water supply.
4. Check the refrigerator and cupboards in your own kitchen. Are all foods covered? Are cleaning products stored away from food? Is the garbage can clean and covered?



MONITORING OUTBREAKS OF FOODBORNE DISEASE

Between 1983 and 1987, a total of 2,397 outbreaks of foodborne disease representing 91,678 cases were reported to the Centers for Disease Control (CDC). The cause could not be determined in 62% of outbreaks, but the following table shows the proportion of outbreaks and cases caused by various agents when the cause was known.

Cause	Outbreaks, %	Cases, %
Bacteria	66	92
Chemicals	26	2
Viruses	5	5
Parasites	4	<1

Salmonella caused 57% of the bacterial disease outbreaks and was the most common pathogen. Fish poisoning due to ciguatoxin and scomrotoxin caused 73% of the outbreaks due to chemical agents. *Trichinella spiralis* caused all parasitic disease outbreaks for 3 of the 5 years; three *Giardia* outbreaks occurred during the other 2 years. Hepatitis A caused 71% of the outbreaks due to viruses.

For each year from 1983 to 1987, the most commonly reported food preparation practice that contributed to foodborne disease was improper storage or holding temperatures, followed by poor personal hygiene of the food handler. Food obtained from an unsafe source was the least commonly reported factor. Inadequate cooking and contaminated equipment each ranked third or fourth in each of the 5 years.

Many more outbreaks occur than are reported to CDC. An outbreak is more likely to be reported if it is easily recognized and the source is confirmed by laboratory analysis of feces, blood, or food. Data on sporadic, individual cases of foodborne disease are not included, but are far more common than outbreaks.

CDC monitors foodborne disease outbreaks for three reasons:

1. *Disease prevention and control.* Surveillance of foodborne disease allows contaminated foods to be removed from stores quickly, shows when faulty food preparation practices need to be corrected in foodservice and at home, and identifies human carriers of foodborne pathogens, who can then be treated.

2. *Knowledge of causes of disease.* The pathogen causing a foodborne disease outbreak is often unidentified, either because it was not tested, the laboratory investigation was too late or incomplete, or the pathogen could not be detected even after thorough testing. Earlier and more thorough investigations and better testing methods are needed.
3. *Administrative guidance.* Data from investigations of foodbourne disease can show emerging problems and common errors in food handling. Surveillance helps to increase the awareness of food protection methods, leads to better training programs, and encourages use of available resources.

Most reports come to CDC from state and local health departments, but some also come from federal agencies such as the Food and Drug Administration (FDA), the U.S. Department of Agriculture (USDA), the U.S. Armed Forces, and occasionally from private physicians.

Source: Bean, N. H., Griffin, P. M., Goulding, J. S., and Ivey, C. B. 1990. Foodborne disease outbreaks, 5-year summary, 1983–1987. *MMWR Surveillance Summaries* 39(SS-1):15.